Proposing Digital Copyright Infringements Enforcement through Internet Court in Indonesia: Fostering the Existing E-Court System

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Abstract. As digital transformation advances, digital copyright violations have increased, costing Indonesia over USD 2.5 billion annually. Despite existing laws, enforcement remains slow and ineffective. The enforcement of digital copyright laws through an Internet Court is increasingly critical. China have pioneered this initiative with notable successes, while Indonesia lags in fully adopting digital judiciary practices. This study examines the mechanisms, technologies, and challenges of Internet Court, with a focus on Indonesia and China. Findings indicate that China's Internet Court has significantly reduced case processing time through AI-assisted adjudication and blockchain-backed evidence verification. Over 90% of disputes handled by China's Internet Courts have been resolved without appeal, demonstrating efficiency. Indonesia's current E-court system lacks similar technological integration, limiting its effectiveness in handling digital copyright cases. Challenges include inadequate legal frameworks, lack of digital forensic capabilities, and insufficient internet infrastructure. The study recommends the regulatory reforms as well as starting the implementation of fully online dispute settlement supported by blockchain-backed evidence verification to strengthen Indonesia's digital judiciary.

1 Introduction

The rapid digital transformation has led to a surge in digital copyright violations, exacerbated by the increased use of online platforms. Indonesia, with 212.9 million internet users in 2023, faces mounting copyright infringement cases, paralleling China's experiences. As seen in **Fig. 1** and **Fig. 2**, In 2020, Indonesia recorded approximately 175.4 million internet users, a number that grew to 212.9 million by 2023. Meanwhile. China, the country with the largest number of internet users in the world, reported over 900 million users in 2018, reaching more than 1.05 billion in 2023 [1–3]. The correlation between internet penetration and rising copyright infringement underscores the urgency of stronger enforcement mechanism. China's progressive establishment of Internet Court since 2017 has provided an efficient framework for addressing digital disputes.

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Despite legislative efforts, Indonesia's e-court system struggles with jurisdictional limitations over digital disputes. Additionally, electronic evidence verification remains a significant challenge due to the absence of comprehensive digital forensic protocols.

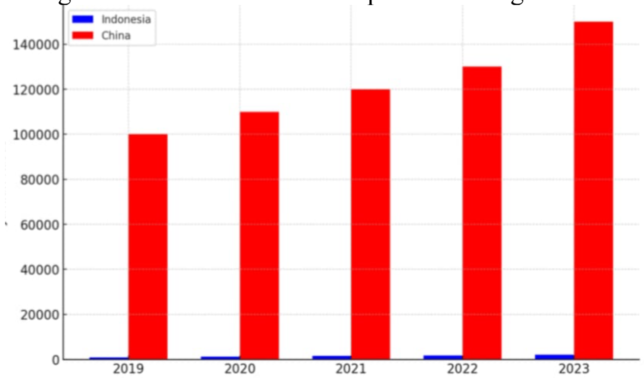

Fig. 1. Internet Users in China and Indonesia

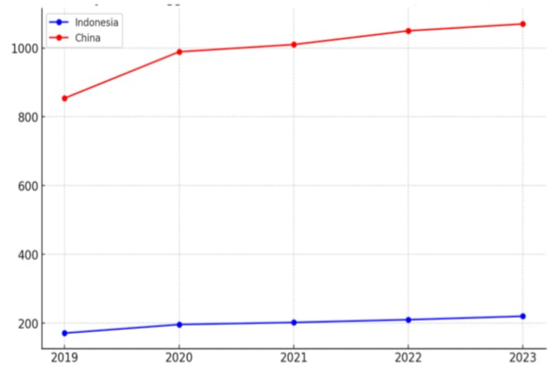

Fig. 2. Digital copyright Violation case in Indonesia and China

This initiative began with the establishment of the Internet Court in Hangzhou on August 18, 2017, followed by similar court in Beijing and Guangzhou in 2018. These courts were designed to address various disputes related to digital activities as well as copyright infringement. On of the key regulations supporting their operations is the Provisions on Several Issues Concerning the Trial of Cases by the Internet Courts, published in 2018. The success of these Internet Courts became evident in 2022, when China's first NFT copyright infringement case, *Shenzen Qice Diechu Cultural Creativity Co. Ltd., vs. Hangzhou Yuanyuzhou Technology Co., Ltd.* resolved [4]. The Hangzhou Internet Court ruled that NFT platforms may be held liable for contributory copyright infringement if they fail to implement IP ownership checks for digital works on their platforms. The case involved a *Fat Tiger* comic image unlawfully sold as an NFT on the Bigverse platform without the copyright owner's consent. The ruling ordered the platform to destroy the infringing NFT and pay damages, emphasizing the importance of IP review mechanisms to maintain trust and legitimacy in the NFT ecosystem. This decision establishes a critical precedent for intermediary liability in China's NFT market. Interestingly, the case was resolved fully online, without any presence of the disputing parties physically, including the process of verification of proof.

Indonesia, on the other hand, facing several digital copyright disputes, one of which, the composer of popular song Harta Berharga, the theme song of TV series Keluarga Cemara. Due to the emergence of digital platforms in the internet, his works were uploaded without his consent for commercial purposes and so on [5]. However, the case was not resolved before the Commercial Court. Indonesia's E-court system under the Supreme Court, falls short of the comprehensive digital capabilities observed in China. Despite legislative efforts, significant gaps remain in infrastructure, regulation, and implementation. This study aims to analyze the existing models, assess challenges, and propose strategies for Indonesia to adopt a robust Internet Court system.

2 Method

This study employs a comparative analysis of Indonesia's and China's digital judiciary systems. It reviews existing literature, legal frameworks, and case studies, focusing on mechanisms, technological adoption (AI, blockchain, and digital forensics), and legal outcomes. Secondary data serves as the primary source for this study.

3 Result and Discussion

3.1 Mechanism of Internet Court in China

China's Internet Court operate entirely online. It handles cases related to digital copyright, e-commerce, and internet-related disputes. Technically, these Internet Courts utilizes digital litigation platforms that enable parties to submit their case online and applications for enforcement, which significantly cutting down the time usually spent in traditional court lines. Additionally, parties involved in legal proceedings can participate in hearings and other processes remotely from their homes or workplaces [6,7]. China's Internet Court leverage Artificial Intelligence (AI) technology to assist in decision-making processes. AI is utilized to support document processing, identify legal patterns, and provide recommendations based on relevant legal precedent. Notably AI is not the sole arbiter but supports human judges in streamlining cases. Most decisions resolved by Internet Court assisted by AI have been accepted without appeal, emphasizing the critical role of AI in expediting court proceedings and enhancing overall efficiency. With a strong emphasis on user consent, meaning parties actively choose to litigate online and are often satisfied with the streamlined process. Additionally, the courts often utilize technology to provide detailed explanations for their decisions, which can reduce the desire to contest the rulings [8].

Lastly, China's Internet Court accept electronic evidence stored on blockchain as legally valid, recognizing this technology as a secure and immutable method for storing and verifying evidence [9]. Blockchain utilizing smart contracts to secure the process of evidence collection, storage, and verification, ensures the authenticity of electronic evidence remain untampered throughout its lifecycle [10,11].

All electronic data used as evidence viewed as meeting the formal requirement of the original. However, in case the defendant objected the authenticity of the evidence, the Internet Courts will review and assess the authenticity of the electronic data. Furthermore, in the Article 11 of The Supreme People's Court's Provisions on Several Issues Related to Trial of cases by the Internet Courts 2018, governed the use of blockchain-backed evidence by the Internet Court as follow: "*...Where the authenticity of electronic data submitted by parties can be proven through electronic signatures, reliable time stamps, hash value checks, blockchain, and other technological methods for collecting, fixing, and preventing alteartion of evidence, or verified through the electronic evidence collection and storage platform, the Internet Courts shall confirm it..*". Means that blockchain technology is primarily utilized through third-party platforms. The court does not operate directly the blockchain infrastructurem but insetad relies on the external services to store and verify blockchain-based evidence, e.g, the Beijing Zhongjing Tianping platform [12,13].

Fig. 3 below illustrates the litigation process through the Internet Court, along with the technologies utilized. Prior to entering this process, the parties involved are facilitated through pre-trial mediation. If the mediation fails, the case will proceed to the litigation process.

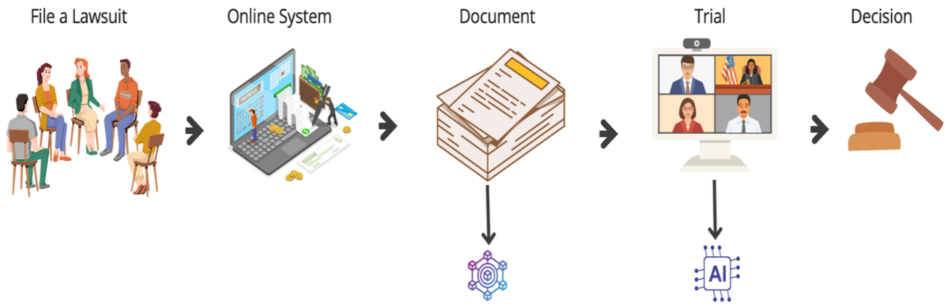

Fig. 3. China's Internet Court Mechanism

As such, China's Internet Courts provide a flexible and efficient framework for addressing modern disputes, particularly those involving technology and the internet. This technology not only enhances access to justice but also accelerates court proceedings and enables more adaptive law enforcement in the digital era.

3.2 Indonesia's E-court: Limitations

Indonesia's E-court system, governed by Supreme Court Number 7 of 2022 concerning Electronic Case Administration and Court Proceedings, facilitates electronics case filing and fee payment. The introduction of the Electronic Court (E-court) system is also part of the implementation of the Electronic-Based Government System (SPBE). The aims of the e-court establishment is expected to be a breakthrough in judicial digitalization, encompassing case document submission, payment of court fees, electronics summons, and electronic trials, as well as submissions of evidence. However, in practice, the E-court system has not been fully and effectively implemented. Most hearings still require physical attendance, including during the examination of proof and testimony of witnesses and or experts in the digital copyright disputes. In Indonesia, the recognition of electronic evidence is already regulated under Law on Electronic Information and Transaction (EIT) Number 1 of 2024 on the Second Amendment of Law No. 11 of 2008 on EIT. In practice, challenges arise in verifying evidence that requires digital forensics. The evidence verification system in Internet Courts must be equipped with strict standards and protocols. Digital forensics should become an integral part of this process, utilizing tools that can authenticate the validity of digital evidence, such as metadata trails and digital certificates supported by blockchain technology. Unfortunately, Indonesia does not yet have the technological infrastructure and regulatory framework necessary to ensure the authenticity of submitted evidence. In practice, digital forensics experts were hired to analyze metadata. However, the lack of standardized protocols led to disputes over the admissibility of the findings, ultimately delaying the resolution in the case. In addition, it would lead to additional cost for disputing parties using such traditional method by hiring the expert. Indonesia lacks such regulations, leading to a higher risk of rejection or skepticism regarding digital evidence in court. Thereby, limiting the system's ability to provide full flexibility in accessing justice digitally.

Another major obstacle in implementing Internet Courts in Indonesia is the uneven distribution of digital infrastructure. While Indonesia boasts one of the largest digital markets in Southeast Asia, with over 200 million internet users, technological infrastructure particularly in rural areas remains inadequate to support fully digital judicial processes. According to data from the Central Bureau of Statistics (BPS) in 2023, around 20% of regions in Indonesia still lack stable internet access, especially in the eastern parts of the country [14].

This situation hampers the smooth execution of digital court proceedings, as not all courts have access to reliable infrastructure, such as high-speed internet and secure data processing servers. Advanced technologies like video conferencing and digital case management systems require dependable infrastructure, and without it, hearings can be disrupted or even canceled due to network issues. In comparison, countries like China benefit from robust and well-distributed technological infrastructure to support their Internet Courts, while Indonesia is still in its early stages in this regard.

Beyond technological challenges, human resources in Indonesia also pose a significant hurdle in the implementation of Internet Courts. The use of advanced technology in the judicial system such as artificial intelligence (AI), blockchain, and digital forensics demands a high level of technical expertise from judges, prosecutors, lawyers, and court staff. However, in practice, many still struggle to understand and utilize digital tools effectively. According to the 2023 World Bank report, Indonesia's digital literacy rates remain low compared to neighboring countries in Asia, such as Singapore and Malaysia [15].

In addition to the Supreme Court Regulation on E-court, Indonesia has also introduced an online arbitration system through the 2022 regulations issued by the Indonesian National Arbitration Board (BANI). These regulations provide technical guidelines for conducting arbitration processes online, including the examination of written evidence and the testimony of witnesses and experts via video conferencing facilities. However, the legal option of arbitration has its drawbacks, as resolution through arbitration requires the inclusion of an arbitration clause either in the agreement made prior to the dispute or after the dispute has occurred. This means that if one party refuses to agree to arbitration, it become challenging for the parties to resolve the dispute through this method.

The implementation of E-court in Indonesia, while a significant step forward in modernizing the judicial system, is still far from optimal. Currently, the system does not fully address the increasingly complex demand of the digital era, especially in disputes involving the internet and digital technologies. There is an urgency for more specific and comprehensive regulations, particularly those issued by the Supreme Court through Supreme Regulations, to provide detailed guidance on Internet Court in Indonesia. On top of that, Indonesia must put concern on the status of blockchain technology in its regulation. No current laws in Indonesia explicitly mentioned the blockchain technology. Even though in practice, the use of blockchain technology through the implementation of crypto assets on commodity futures exchanges clearly uses blockchain technology (Regulation of the Commodity Futures Trading Supervisory Agents/BAPPEBTI Regulation).

While China's Internet Court demonstrate scalability and efficiency, Indonesia faces hurdles in technological readiness, regulatory gaps, and digital literacy among judiciary stakeholders. Addressing these issues requires targeted reforms.

3.3 Proposed Framework for Indonesia

The regulation of Civil procedural Law in Indonesia is currently a priority in the national legislative program, presenting opportunity to adopt rules for dispute resolution using technology. Furthermore, the technical implementation of dispute resolution through e-court particularly regarding the use of blockchain for evidence verification and integration of AI into judicial processes could be governed by Supreme Court Regulation. Thus, beside enabling full digital litigation for Copyright infringement settlement, it also recognized blockchain-stored evidence as legally binding. Virtual hearing is not entirely new in

Indonesia, as it was implemented during the Covid-19 pandemic. Nevertheless, it was limited to criminal court proceedings and have not been fully extended to civil or other types of cases. In criminal court proceeding, the virtual hearings were used to improve efficiency and case management [16].

Furthermore, strengthening digital infrastructure is essential. As Internet Court require stable internet connectivity, the government must ensure that internet infrastructure across Indonesia can effectively support the conduct of digital court proceedings. The development of digital judiciary requires input and support from private sector entities, including tech companies and Internet Service Providers (ISP). Private companies can provide technological expertise, cybersecurity measures, and funding to develop secure digital platforms. This partnership ensures that the judicial system remains updated with the latest technological advancements and has access to innovative solutions for digital evidence verification, AI-driven case management, and blockchain security. Additionally, capacity building e.g train the judges and legal practitioners by The Supreme Court-a body authorized to provide judges through periodic training-on digital tools as well as enhance awareness of digital judiciary is also essential to realizing effective law enforcement through Internet Courts. As the digital landscape evolves, so too should the judicial system. The introduction of AI and blockchain in legal processes requires constant monitoring to ensure that these technologies remain relevant and effective. The establishment of a mechanism for continuous improvement and feedback from users of the E-court system is needed for an ongoing evaluation and adaptation process.

4 Conclusion

The development of Internet Courts presents significant potential for resolving modern disputes, particularly those involving digital copyright and internet-related issues. While China has successfully implemented a comprehensive Internet Court system with advanced technologies such as blockchain and artificial intelligence, Indonesia's E-Court system remains in its early stages, facing challenges in infrastructure, regulation, and digital literacy. To address these gaps, targeted reforms are necessary, including the creation of detailed Supreme Court regulations to govern Internet Courts, recognize blockchain-stored evidence as legally binding, and integrate AI into judicial processes. Strengthening digital infrastructure, ensuring stable internet connectivity nationwide, and providing capacity-building initiatives to train judges and legal practitioners in digital tools are also critical steps. By implementing these measures, Indonesia can establish an efficient Internet Court system that aligns with the demands of the digital era, enhances access to justice, and modernizes its judicial processes.

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